Self-Supervised Link Prediction vs. Supervised Classification in Graph Neural Networks for Anti-Money Laundering

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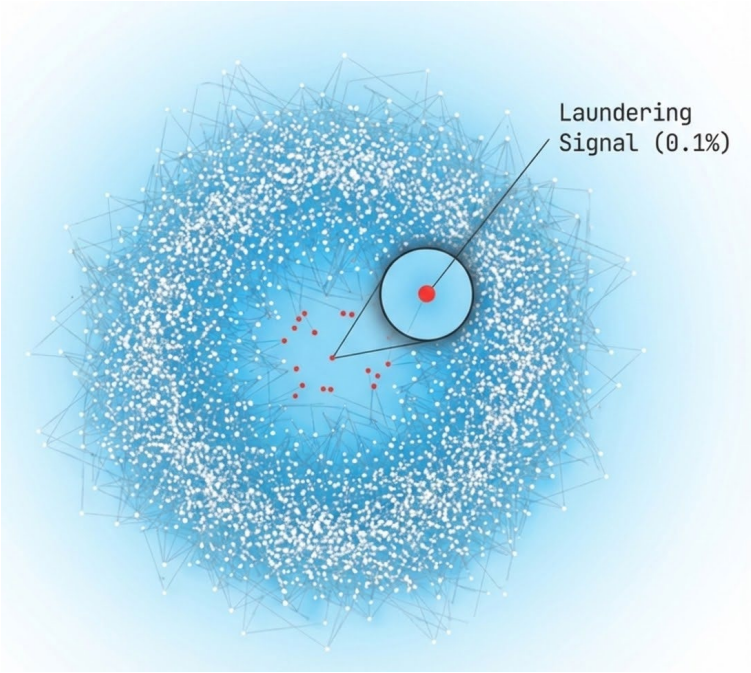
Feb 2026

Introduction

- **Economic Impact:** Global money laundering accounts for 2–5% of GDP (€1.7 T – €4T), yet detection remains inefficient.
- **False Alarms:** Current rule-based systems suffer from >95% False Positive Rates, burying real threats in noise.
- **Label Scarcity:** Real-world AML labels are rare, delayed, and expensive, making standard supervised learning impractical.
- **The Opportunity:** Financial data is inherently relational. Graphs allow us to move beyond transaction "snapshots" to full structural context.



Methodology: LaundroGraph

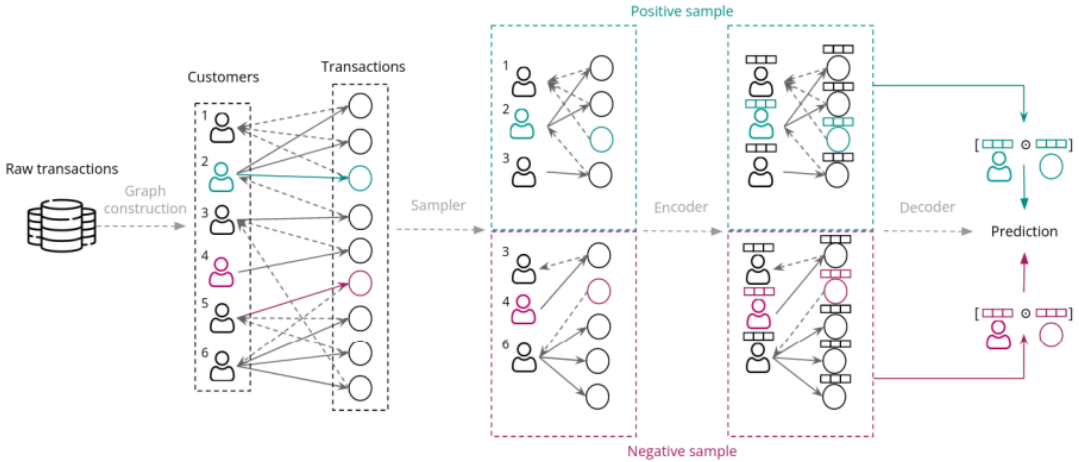


Figure 1: Proposed system training overview. Outgoing transactions are represented with filled arrows, and incoming transactions with dashed arrows. First, the bipartite graph is built from a dataset comprised of raw transactions. Then, positive pairs (green) and negative pairs (red) together with their K-hop subgraphs ($K = 2$ in the figure) are extracted and their embeddings obtained through the encoder. Finally, the decoder uses the aforementioned embeddings to generate the prediction for each sampled edge.

- **Graph Design:** A **Directed Bipartite Graph** $\boldsymbol{G} = (\boldsymbol{C} \cup \boldsymbol{T}, \boldsymbol{E})$, linking Customers (C) and Transactions (T).
- **Learning Task:** Self-Supervised **Link Prediction** via graph reconstruction (**no labels required**).
- **Encoder-Decoder Architecture:**
  - **Encoder:** Aggregates neighborhood signals via **Graph Attention (GAT)** to project nodes into a shared latent space $(\boldsymbol{z_c}, \boldsymbol{z_t} \in \mathbb{R}^{\boldsymbol{d}})$.
  - **Decoder:** Interaction function $\boldsymbol{s} = \boldsymbol{\sigma}(\boldsymbol{W}[\boldsymbol{z_c} \odot \boldsymbol{z_t}])$ using the **Hadamard** product to score link compatibility.
- **The Logic:** Link with low predicted probability is flagged as **Structural Anomaly** (Potential Money Laundering).

Dataset & Feature Engineering (SAML-D)

- **Composition: 855k Customers** and **9.5M Transactions**.
- **Typologies**: 28 distinct behavioral patterns (17 suspicious / 11 normal)
- **Temporal Split:**
  - **Profile Window (2022/10 – 2023/02):** Customer Profiling Period.
  - **Training Window (2023/03 – 2023/07):** Model learning phase.
  - **Testing Window (2023/08):** Out-of-time evaluation on unseen future transactions.
- **Laundering Rate (Target):** 0.1%.

Customer Nodes (123 Dimensions):

- **Behavioral Aggregates:** Historical transaction volume, frequency (velocity), In/Out fund balance, cash-volume ratios, avg. "Cash-to-Wire" speed, rapid "In-and-Out" counts.
- **Anomaly Indicators:** Engineered counts for "Structuring" behavior
- **Risk Context:** Density of interactions with high-risk jurisdictions or sanctioned entities.

Transaction Nodes (49 Dimensions):

- **Intrinsic Properties:** Numerical amount (log-normalized), currency types, payment channels, bank locations.
- **Temporal Dynamics**: Cyclic time features (Hour/Day) and "Latency" (the speed at which funds move from cash-to-wire).

Time	Date	Sender_account	Receiver_account	Amount	Payment_currency	Received_currency	Sender_bank_location	Receiver_bank_location	Payment_type	Is_laundering	Laundering_type
10:54:03	2023-06-15	4368387605	1475956965	15322.40	UK pounds	UK pounds	UK	UK	ACH	0	Normal_Fan_Out
11:18:58	2023-02-10	6396278172	5890679423	76283.69	UK pounds	UK pounds	UK	UK	ACH	0	Normal_Mutual
09:32:25	2022-11-28	3717802726	7803261738	21204.72	UK pounds	Turkish lira	UK	Turkey	Cross-border	0	Normal_Small_Fan_Out
21:55:49	2023-02-17	9318946892	3606432667	11718.68	UK pounds	UK pounds	UK	UK	Credit card	0	Normal_Fan_In
20:32:39	2022-10-27	12470473	7705228058	1422.56	UK pounds	UK pounds	UK	UK	Credit card	0	Normal_Small_Fan_Out

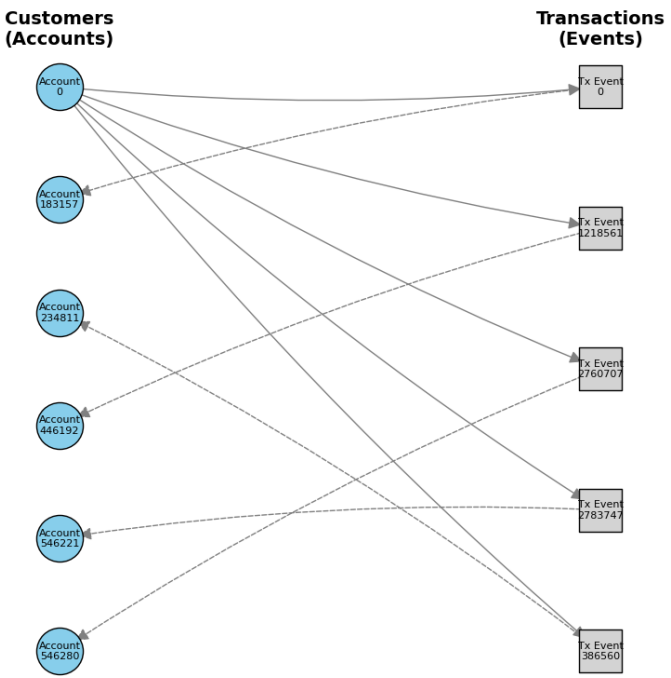
Graph Construction

Node Counts:

- **Total Customers:** 0.7M
- **Total Transactions:** 5.2M
- **Graph Ratio:** 1 Customer ≈ 8 Transactions

Edge Splits:

- **Training Edges:** 4.5M
  - **Message Passing:** 2.3M (50%)
  - **Supervision:** 1.4M (30%)
  - **Validation:** 0.9M (20%)
- **Test Edges:** 0.7M



Experiment 1: LaundroGraph (Self-Supervised approach)

Encoder:
→ *Linear projection (64)*
→ *GAT (2 layers, 2 heads)*
→ *Residual (self + neighbors)*
= *Node embeddings* $(z_c, z_t) \in \mathbb{R}^{64}$

Decoder:

→ $z_e = z_c \odot z_t \in \mathbb{R}^{64}$

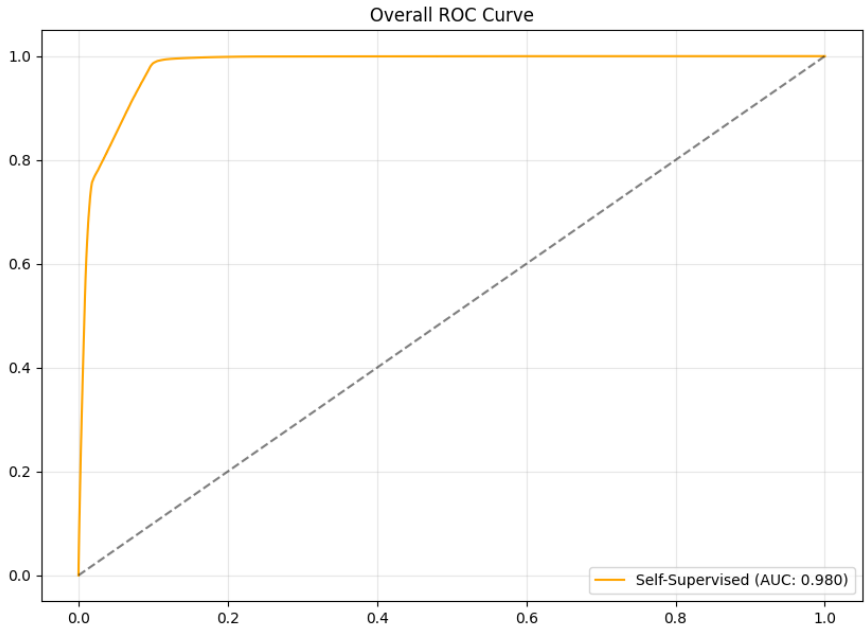
→ $MLP(z_e)$ *(64 → 32 → 1)*

= $Score\left(y_{c,t}^{link} = 1 \middle| z_c, z_t\right)$, *where* $y_{c,t}^{link} \in \{0,1\}$

$$\Rightarrow Anomaly\ Score = 1 - Score\left(y_{c,t}^{link} = 1 \middle| z_c, z_t\right)$$

Training Objective:
→ The model optimizes ***Binary Cross-Entropy (BCE)*** Loss to distinguish between observed edges ($\boldsymbol{E}^+$) and negative sampled edges ($\boldsymbol{E}^-$) with one negative edge sampled per positive edge (M=1)

Training Hyperparameters:
→ *Learning Rate: 0.001*
→ *Neighbors Sampling: 1-hop 20; 2-hop 10;*
→ *Batch Size: 2048*



Experiment 2: Edge Classification (Supervised approach)

Encoder:
→ *Linear projection (64)*
→ *GAT (2 layers, 2 heads)*
→ *Residual (self + neighbors)*
→ *LayerNorm*
= *Node embeddings* $(z_c, z_t) \in \mathbb{R}^{64}$

Decoder:

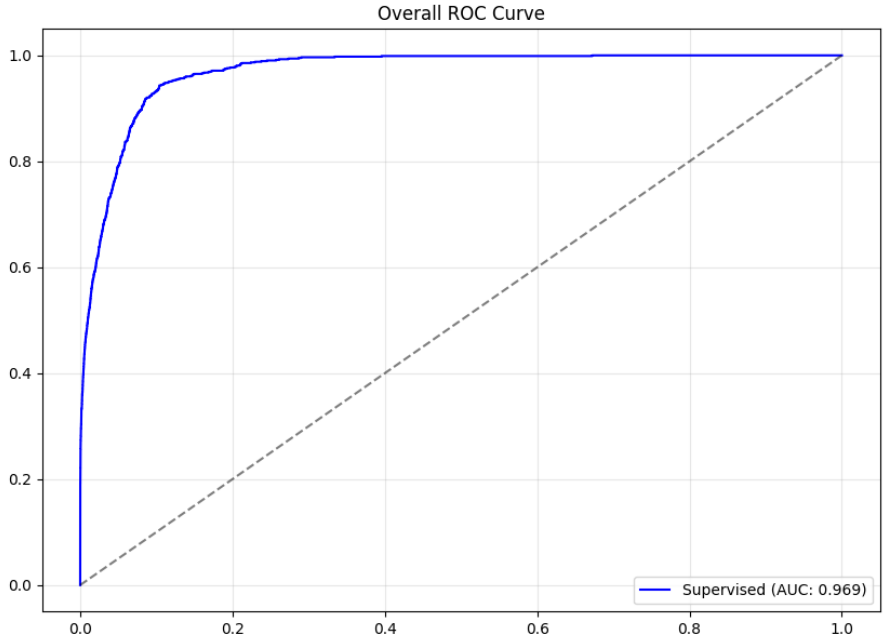
→ $z_e = z_c \parallel z_t \in \mathbb{R}^{128}$

→ $MLP(z_e)$ *(128 → 32 → 1)*

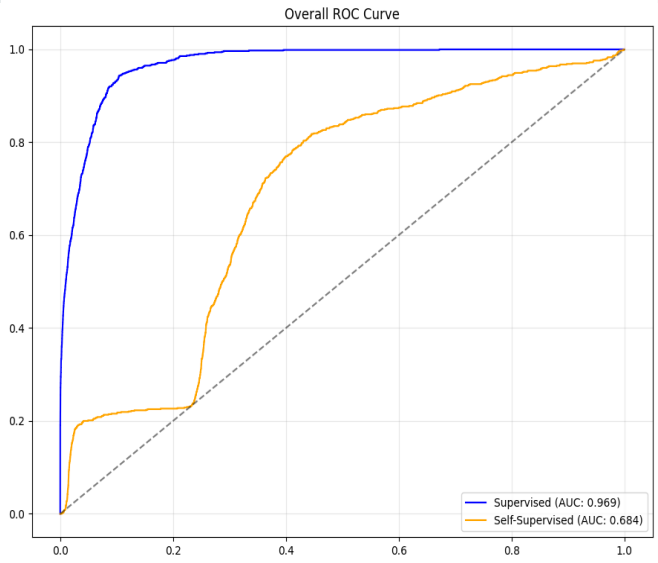
= $Score\left(y_{c,t}^{ML} = 1 \mid z_c, z_t\right),\ where\ y_{c,t}^{ML} \in \{0,1\}$

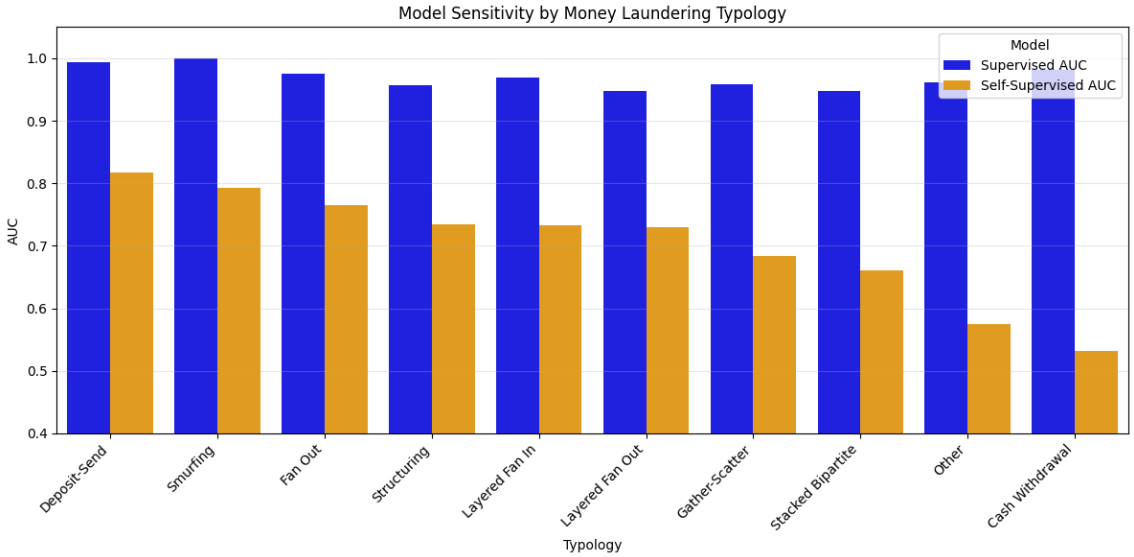
Training Objective:
→ The model is trained using a **weighted Binary Cross-Entropy (BCE)** loss for edge-level laundering classification.
→ $Weight = \frac{Negatives}{Postivies} = 939.8$

Training Hyperparameters:
→ *Learning Rate: 0.001*
→ *Neighbors Sampling: 1-hop 20; 2-hop 10;*
→ *Batch Size: 1024*



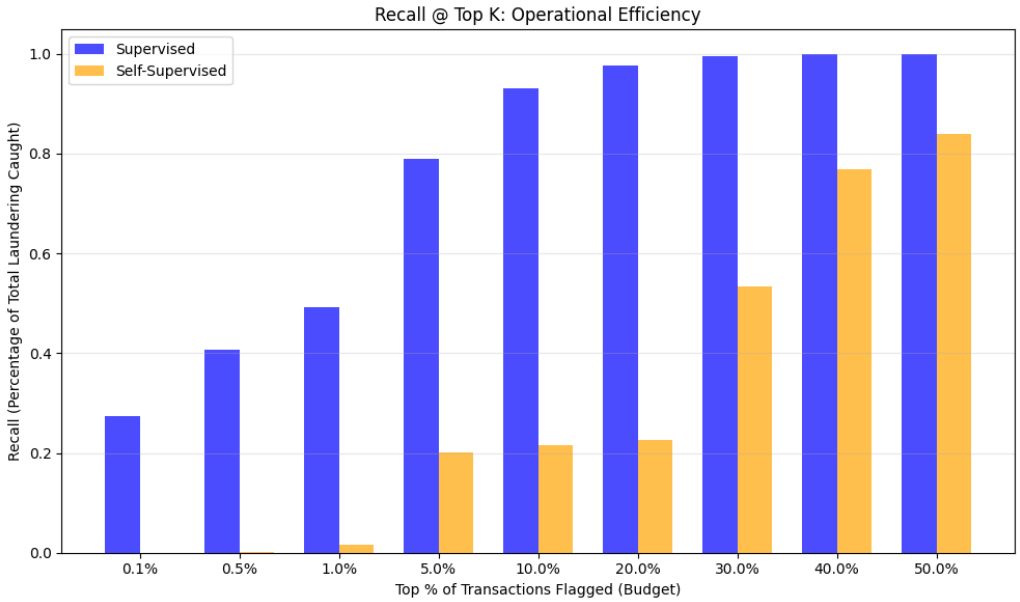
Self-Supervised VS Supervised approaches





- **Performance Insight:** While Supervised learning dominates overall, the Self-Supervised model achieves strong detection (~80% AUC) on some crime type (like Smurfing) without using any labels.

Recall @ K (Operational Effectiveness)



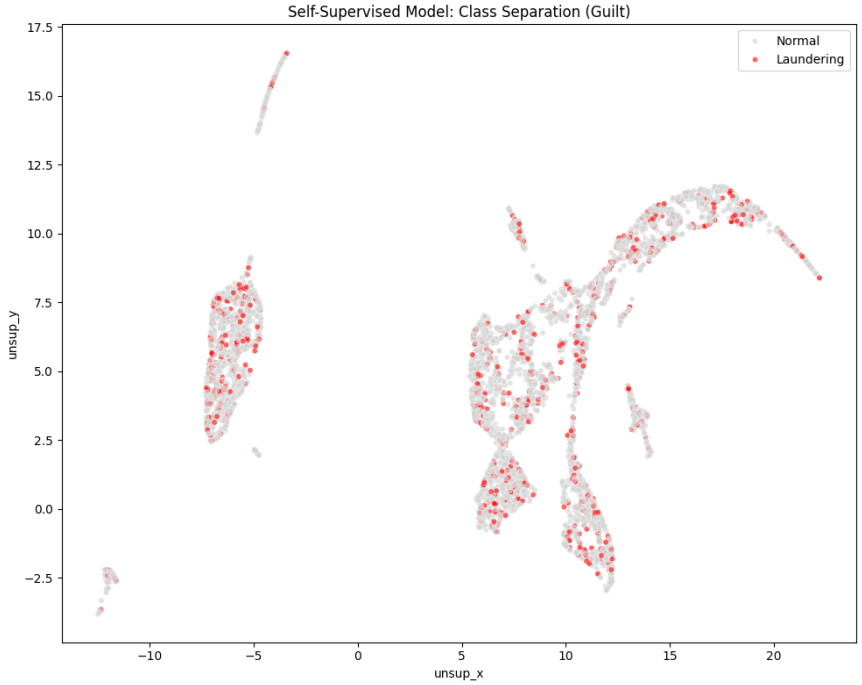
Supervised:

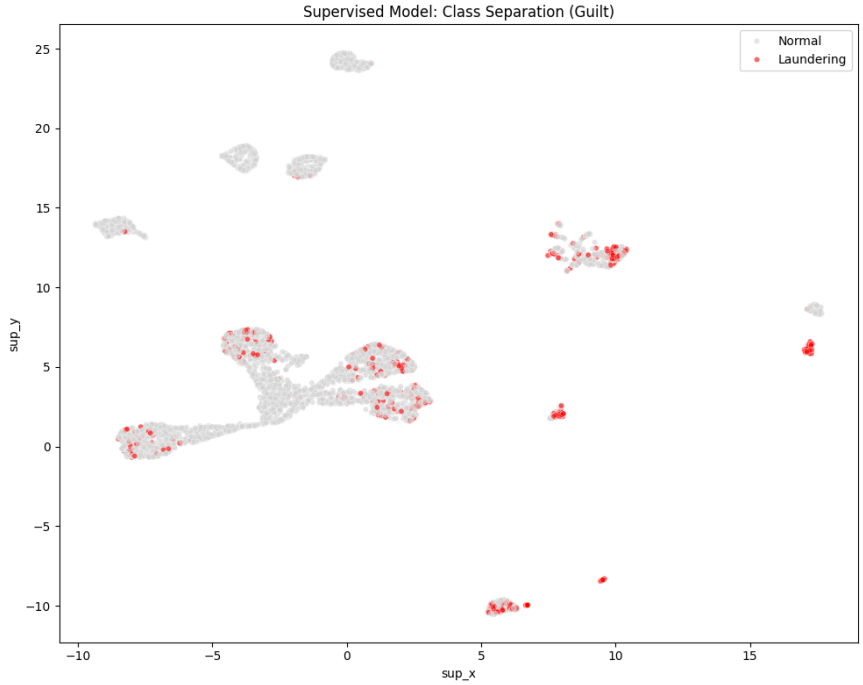
- **High Efficiency:** Catches **~80% of money laundering** by reviewing just the top **5%** of transactions.
- **Operational Role:** Ideal for primary, automated blocking of known threats.

Self-Supervised:

- **Lower Efficiency:** Requires a larger review budget (top 30-40%) to achieve comparable recall**.**
- **Why?** It flags all anomalies, not just fraud. The top 1% contains many "weird but legal" transactions (False Positives).
- **Operational Role:** Best used as a secondary **"Safety Net"** to investigate accounts that pass the Supervised check but behave strangely.

Embeddings (Transactions)





- **Result:** Significant overlap. The Red points are mixed deep within the Grey clusters.
- **Reason:** The model was never shown labels. It groups transactions based on structural similarity.

- **Result:** Perfect separation. The "Laundering" (Red) points are pushed far away from "Normal" (Grey) points.
- **Reason:** The model was explicitly trained to discriminate between these two classes. It learned "Guilt."

Conclusions

Geometry Drives Detection:

- **Success (Smurfing, ~80% AUC):** The model excels at detecting crimes with unnatural shapes. Typologies like Smurfing create "Star" patterns (high-degree hubs) that stand out mathematically as structural anomalies
- **Limitation (Cash Withdrawal, ~53% AUC):** Crimes that lack a distinct shape fail. A withdrawal creates a Single Edge, which looks topologically identical to a legitimate purchase.

The "Structural Mimicry" Barrier:

- Self-Supervised learning struggles when criminal behavior intentionally **mimics** complex legal flows (e.g., **Stacked Bipartite** resembling payroll). Without labels, the model cannot separate these "good" and "bad" twins.

A Hybrid Strategy:

- **Supervised Models:** Deploy as the "Front Line" to block known, simple threats efficiently
- **Self-Supervised Models**: Deploy as a **"Discovery Tool"** to uncover rare, complex anomalies that supervised rules have never seen before.

References

1. Mário Cardoso et al. (2022) “LaundroGraph: Self-Supervised Graph Representation Learning for Anti-Money Laundering”

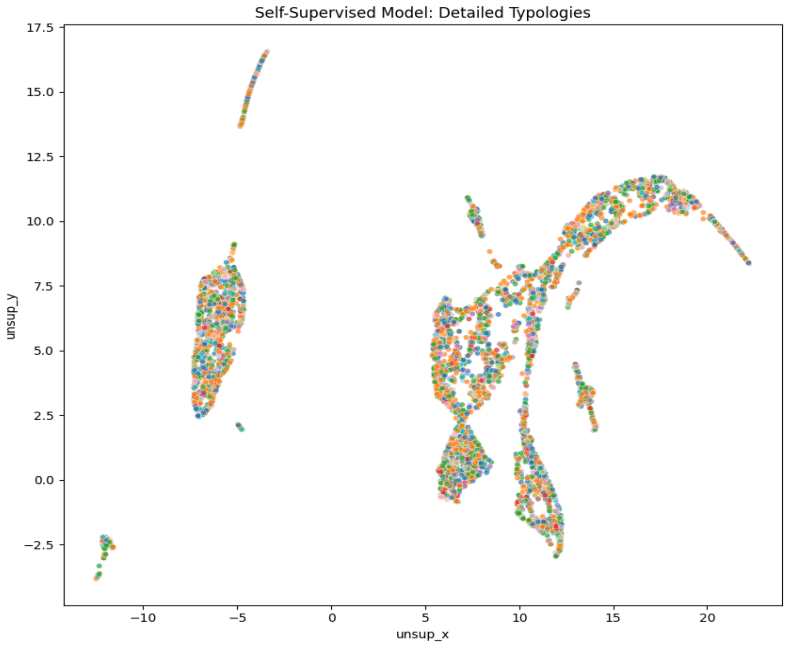
2. Berkan Oztas et al. (2023) “Enhancing Anti-Money Laundering: Development of a Synthetic Transaction Monitoring Dataset”

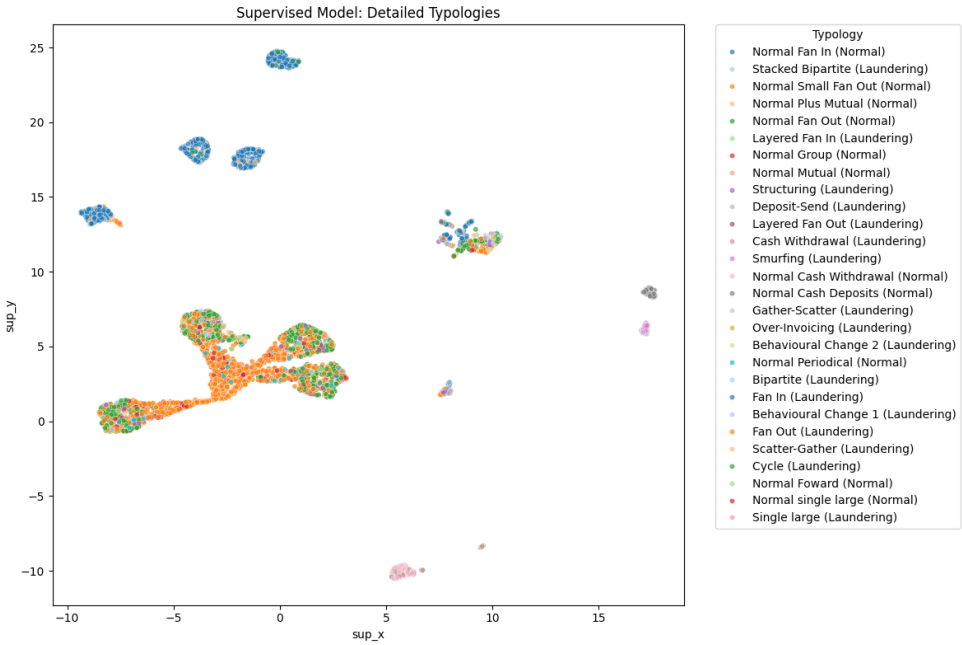
3. Dawei Cheng et al. (2024) “Graph Neural Networks for Financial Fraud Detection: A Review“

Thank you for the attention!

Feb 2026

Extra Slides (Embeddings)





Extra Slide (LaundroGraph Algorithm)

Algorithm 1 LaundroGraph forward propagation algorithm

Input: Graph G; number of layers L; neighborhood sampler $\mathcal{N}$; mini-batch size B; edge sampling function $\mathcal{S}$; edge direction D

$E_p : (c_1, t_1), ..., (c_B, t_B) \leftarrow$ select B edges from G in direction D
$E_n : (\tilde{c}_1, \tilde{t}_1), ..., (\tilde{c}_B, \tilde{t}_B) \leftarrow \mathcal{S}(G)$ ▹ Sample random c and t as non-edges
$E \leftarrow E_p \cup E_n$
if D == outgoing **then**
  $G \leftarrow G \setminus (c \rightarrow t), \forall t \in E$ ▹ Delete real outgoing edges
else
  $G \leftarrow G \setminus (t \rightarrow c), \forall t \in E$ ▹ Delete real incoming edges
end if
$z^0_{c_i}, z^0_{t_i} \leftarrow f_{c_i}, f_{t_i}, \forall (c_i, t_i) \in (N_{(c)} \cup c, N_{(t)} \cup t), \forall (c, t) \in E$ ▹ Input to the first layer is the raw features of all required nodes
for $l \in 1, ...L$ **do**
  for $(c, t) \in E$ **do**
    $z^l_c \leftarrow \text{Convolve}(\{z^{l-1}_{c_i}, \forall c_i \in \mathcal{N}_{(c)} \cup c\})$ ▹ Encode nodes
    $z^l_t \leftarrow \text{Convolve}(\{z^{l-1}_{t_i}, \forall t_i \in \mathcal{N}_{(t)} \cup t\})$ ▹ Encode nodes
  end for
end for
$\hat{y}_{c,t} \leftarrow \sigma\left(W[z^L_c \odot z^L_t]\right), \forall (c, t) \in E$ ▹ Decoder edge prediction